Course Topics

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1 BUS 411 — Fixed Income

1.1 Lecture Topics

This document lists the planned lecture topics and subtopics for the course.

1.2 Lecture 1: History and Role of Fixed Income Markets

1.2.1 1.1 What is fixed income?

- Definition of fixed income securities
- Debt versus equity financing
- Main issuers and investors

1.2.2 1.2 Historical development of fixed income markets

- Early sovereign borrowing
- Development of government debt markets
- Growth of corporate bond markets

1.2.3 1.3 Economic role of fixed income securities

- Funding governments
- Funding corporations
- Saving, retirement, and liability management

1.3 Lecture 2: Time Value of Money and Discounting

1.3.1 2.1 Time value of money

- Present value
- Future value
- Discount factors

1.3.2 2.2 Compounding conventions

- Annual compounding
- Semiannual compounding
- Continuous compounding

1.3.3 2.3 Discounting fixed income cash flows

- Cash flow timelines
- Pricing as present value of future cash flows
- Relation between discount rates and value

1.4 Lecture 3: Annuities and Liability Valuation

1.4.1 3.1 Fixed annuities

- Ordinary annuities
- Annuities due
- Perpetuities

1.4.2 3.2 Variable annuities

- Basic structure
- Investment-linked cash flows
- Valuation considerations

1.4.3 3.3 Defined benefit pension plans

- Nature of pension liabilities
- Discounting promised payments
- Funding and valuation issues

1.4.4 3.4 Liability valuation

- Present value of liabilities
- Matching assets to liabilities
- Liability sensitivity to interest rates

1.5 Lecture 4: Government Bonds

1.5.1 4.1 Types of government securities

- Treasury bills
- Treasury notes
- Treasury bonds

1.5.2 4.2 Features of government bonds

- Face value
- Coupon rate
- Maturity
- Clean price and dirty price

1.5.3 4.3 Valuation of government bonds

- Pricing from discounted cash flows
- Yield to maturity
- Accrued interest

1.5.4 4.4 Government bond markets

- Primary market
- Secondary market
- Benchmark role of sovereign debt

1.6 Lecture 5: Corporate Bonds and Credit Risk

1.6.1 5.1 Corporate bond features

- Secured and unsecured bonds
- Seniority
- Covenants

1.6.2 5.2 Valuation of corporate bonds

- Pricing with promised cash flows
- Credit spread over government rates
- Yield spread interpretation

1.6.3 5.3 Credit risk

- Default risk
- Recovery risk
- Credit ratings

1.6.4 5.4 Corporate bond market considerations

- Liquidity
- Issuer quality
- Term to maturity and risk

1.7 Lecture 6: Yield Measures and the Term Structure of Interest Rates

1.7.1 6.1 Yield measures

- Current yield
- Yield to maturity
- Holding period return

1.7.2 6.2 Spot and forward rates

- Spot rates
- Forward rates
- Discount function

1.7.3 6.3 Yield curve construction and interpretation

- Normal, flat, and inverted curves
- Economic interpretation
- Expectations about future rates

1.7.4 6.4 Term structure concepts

- Expectations view
- Liquidity preference
- Market segmentation intuition

1.8 Lecture 7: Duration

1.8.1 7.1 Interest rate risk in fixed income

- Bond price and yield relationship
- Why prices fall when yields rise

1.8.2 7.2 Macaulay duration

- Definition
- Weighted average time to cash flow receipt
- Interpretation

1.8.3 7.3 Modified duration

- Definition
- Approximate price sensitivity
- Duration as first-order risk measure

1.8.4 7.4 Determinants of duration

- Coupon
- Maturity
- Yield level

1.9 Lecture 8: Convexity

1.9.1 8.1 Limits of duration-only approximation

- Nonlinearity in bond prices
- Curvature of the price-yield relationship

1.9.2 8.2 Convexity measure

- Definition
- Interpretation
- Why convexity matters

1.9.3 8.3 Duration-convexity approximation

- First-order and second-order effects
- Improved bond price change estimates
- Application to larger yield changes

1.9.4 8.4 Comparing bonds using convexity

- Price sensitivity differences
- Trade-off between yield and convexity
- Portfolio implications

1.10 Lecture 9: Fixed Income Risk Management

1.10.1 9.1 Duration-based risk management

- Measuring portfolio interest rate exposure
- Portfolio duration

1.10.2 9.2 Immunization

- Asset-liability matching
- Duration matching
- Protecting against small parallel shifts in yields

1.10.3 9.3 Convexity in risk management

- Why duration matching alone is not enough
- Role of convexity in immunization
- Rebalancing issues

1.10.4 9.4 Practical limitations

- Non-parallel yield curve shifts
- Model risk
- Transaction costs

1.11 Lecture 10: Bonds with Contingencies

1.11.1 10.1 Bonds with embedded options

- Callable bonds
- Putable bonds
- Convertible features as a general idea

1.11.2 10.2 Valuation issues for contingent bonds

- Cash flows depend on future events
- Option-like features in bond valuation
- Impact on price and yield

1.11.3 10.3 Mortgage-backed and prepayable securities

- Prepayment risk
- Negative convexity
- Valuation challenges

1.11.4 10.4 Risk characteristics

- Reinvestment risk
- Call risk
- Extension and contraction risk

1.12 Lecture 11: Fixed Income Derivative Securities

1.12.1 11.1 Overview of fixed income derivatives

- Why derivatives are used
- Hedging, speculation, and portfolio management

1.12.2 11.2 Interest rate forwards and futures

- Basic contract structure
- Hedging interest rate exposure
- Relation to underlying fixed income securities

1.12.3 11.3 Interest rate swaps

- Fixed-for-floating exchange
- Basic valuation intuition
- Uses in risk management

1.12.4 11.4 Options on interest rates

- Caps
- Floors
- Basic role in managing rate risk

1.13 Coverage of Core Course Themes

The course is organized around three major themes:

1.13.1 1. History of fixed income securities

- Covered primarily in Lecture 1

1.13.2 2. Valuation methods for fixed income securities

- Fixed and variable annuities: Lecture 3
- Government bonds: Lecture 4
- Corporate bonds: Lecture 5
- Defined benefit pension plans: Lecture 3
- Bonds with contingencies: Lecture 10
- Fixed income derivative securities: Lecture 11

1.13.3 3. Fixed income risk management using duration and convexity

- Duration: Lecture 7
- Convexity: Lecture 8
- Risk management and immunization: Lecture 9